

Passages - The Leverage Space Trading Model

Time is the greatest innovator

Think of leverage as a number between zero and one, a number that if the worst case scenario does manifest, is the percentage of account equity you lose.

Often times you can use gambling situations as a proxy for market situations because gambling illustrations tend to be considerably simpler as they include fewer scenarios and therefore simpler distributions, and the mathematics tend to be very similar, except that market probabilities seem to change with time.

The Kelly criterion is never the point that maximises the growth of capital, except in the abstract case of an infinite number of plays.

To maximise the expected average compound growth, we always end up with the uniform F . F is a function of the length of the game for positive expectation games it starts at 1 (0 in negative expectation games). The value that maximises the arithmetic mean holding period return, diminishes slightly each play and asymptotically approaches the value that maximise the geometric mean holding period return. Since all streams are finite in length, we'll always trade sub-optimally.

The Kelly criterion means to maximise the expected value of the logarithm of capital at each play

When you dilute F , that is to trade at lesser

levels of aggressiveness than the optimal F , you decrease your drawdowns arithmetically, but you also decrease your returns geometrically.

For every profitable point to the left of the peak of the f curve, there is a point corresponding to the same level of profit on the right of the peak, but with considerably more drawdown

Fractional Kelly's, and over-betting Kelly's are strategies that show ignorance toward the nature of the f curve and are both counter-productive in the search for an optimal growth rate

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In the same way that MPT models use the less useful return metric of (arithmetic) average expected return versus the Leverage Space Model using geometric average HPRs for returns, similarly, variance in returns as the risk metric in MPT is supplanted with drawdown as primary risk metric in the Leverage Space Model.

Modern Portfolio Theory simply gives a set of points in N dimensions to be at, when in fact, we are in an $N+1$ -dimensional manifold

Optimal Foraging Theory states that organisms forage in a manner so as to maximize their energy intake per unit of time. Organisms (including primates, such as human beings) behave in such a way as to find, obtain, and consume food containing the most calories while expending the least amount of time (calories) possible in doing so. T